AIAssistedCoding

Assignment –4.3

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Batch:47-b

Lab 4: Advanced Prompt Engineering – Zero-shot, One-shot, and Few-shot Techniques

Task 1: Zero-Shot Prompting – Leap Year Check

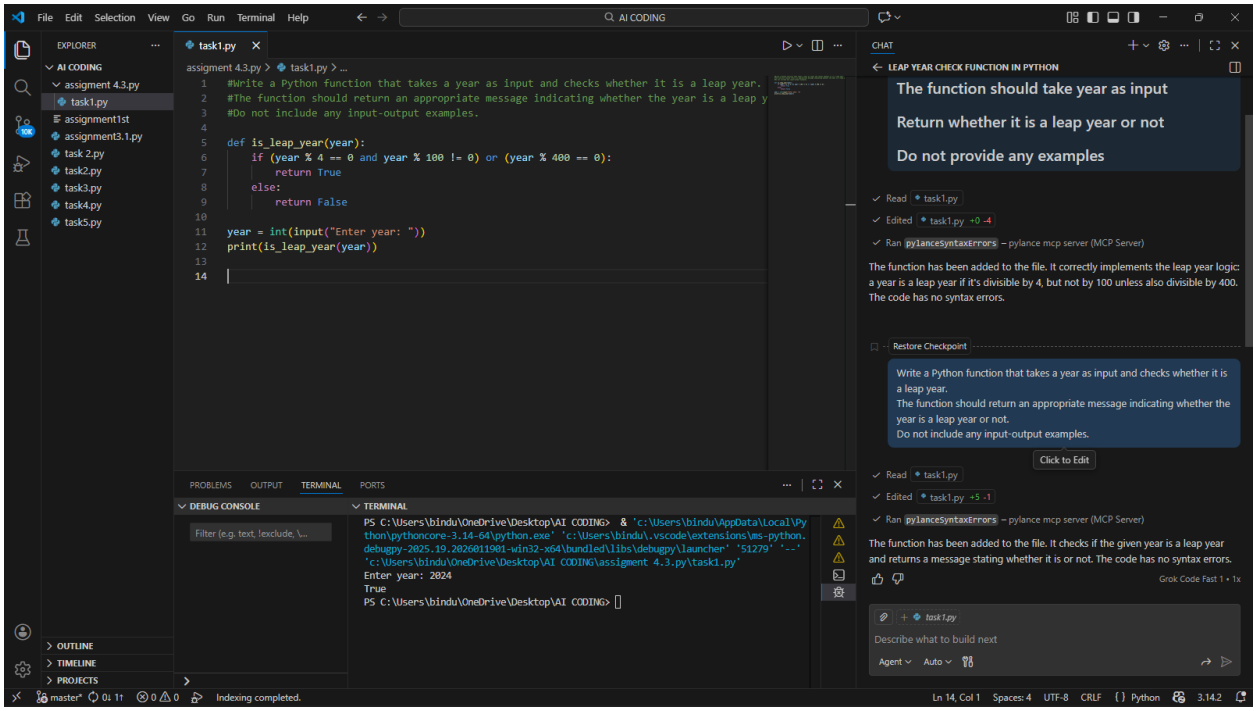
Scenario:
Zero-shot prompting involves giving instructions without providing examples.

Promptused:#Write a Python function that takes a year as input and checks whether it is a leap year.
#The function should return an appropriate message indicating whether the year is a leap year or not.
#Do not include any input-output examples.

- **Generatedcode:**

```
defis_leap_year(year):
    if (year%4==0andyear%100!=0) or (year%400==0):
        returnTrue
    else:
        returnFalse

year=int(input("Enter year: "))
print(is_leap_year(year)
```

- **SampleInput:**
  Enteryear:2024
- **SampleOutput:**
  True
- **ShortExplanationofLogic:**
  the program checks the leap year conditions using logicaloperators .a year is a leap year if the it is divisible by 400 or divisibe by 4 but not divisible by 100 . the logic is implemented directly without providing any prior .



Task2:One-Shot Prompting – Centimeters to Inches Conversion

❖ **Scenario:One-shot prompting guides AI using a single example.**

- **Promptused:**
- # Write a Python function to convert centimeters to inches
- # Use the formula inches = centimeters / 2.54
- # Example:
- # Input: 10 cm
- # Output: 3.94 inches
- **Generatedcode:**

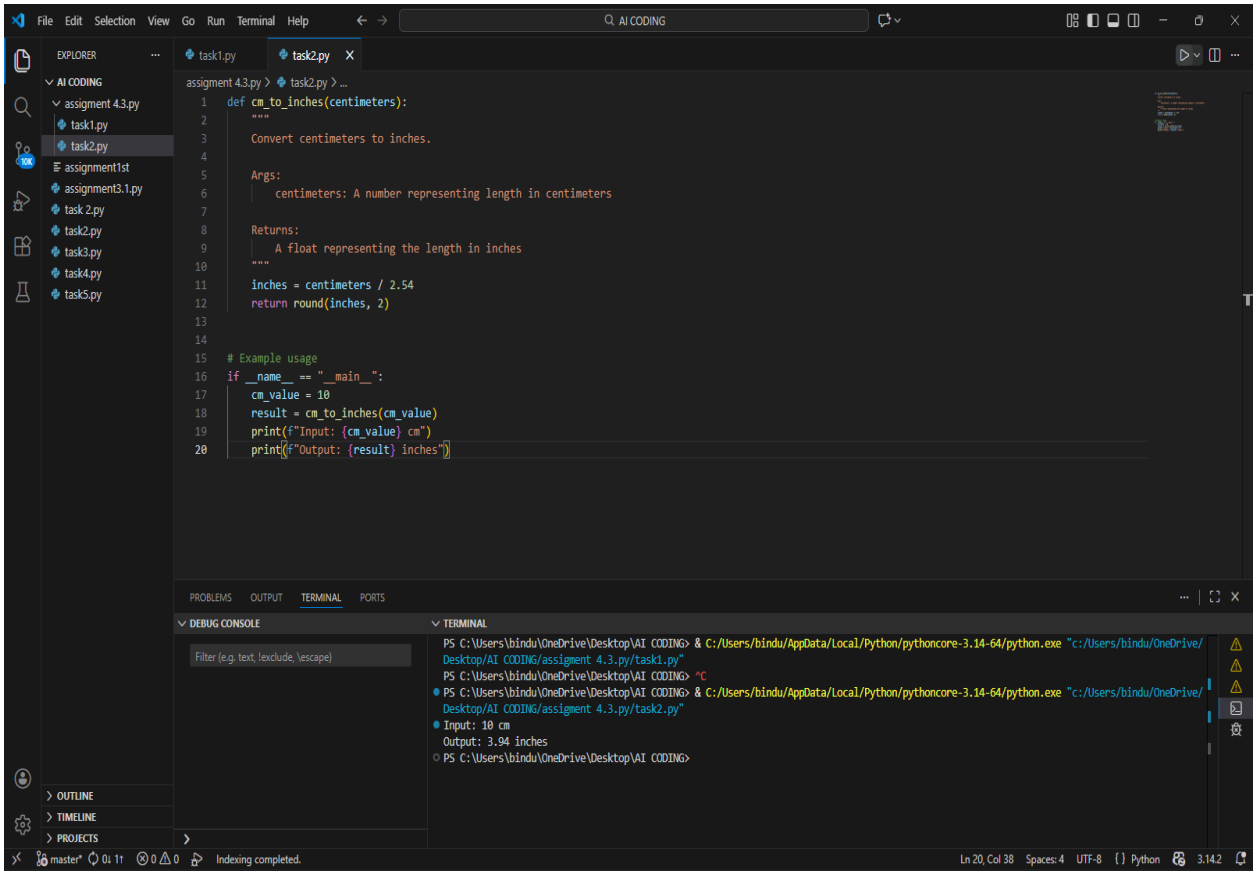
```
defcm_to_inches(centimeters):
    """
    Convert centimeters to inches.

    Args:
        centimeters: A number representing length in centimeters

    Returns:
        A float representing the length in inches
    """
    inches=centimeters/2.54
    returnround(inches, 2)


# Example usage
if__name__=="__main__":
    cm_value=10
    result=cm_to_inches(cm_value)
    print(f"Input: {cm_value} cm")
    print(f"Output: {result} inches")
```

- **SampleInput:**
- Enter value in cm: 25.4
- **SampleOutput:**

  10.0 inches

  ShortExplanationofLogic:

The function takes centimeters as input and converts it into inches using the standard formula Cm %2.54 one example in the prompt helps the ai to generate the logic accurately in one short



Task3:Few-shot promting-Name Formatting

Scenario:

Few-shot prompting improves accuracy by providing multiple examples.

❖

- **Promptused:**
- # Write a Python function to format names as "Last, First"
- # Examples:
- # "John Smith" -> "Smith, John"
- # "Anita Rao" -> "Rao, Anita"

- **Generatedcode:**

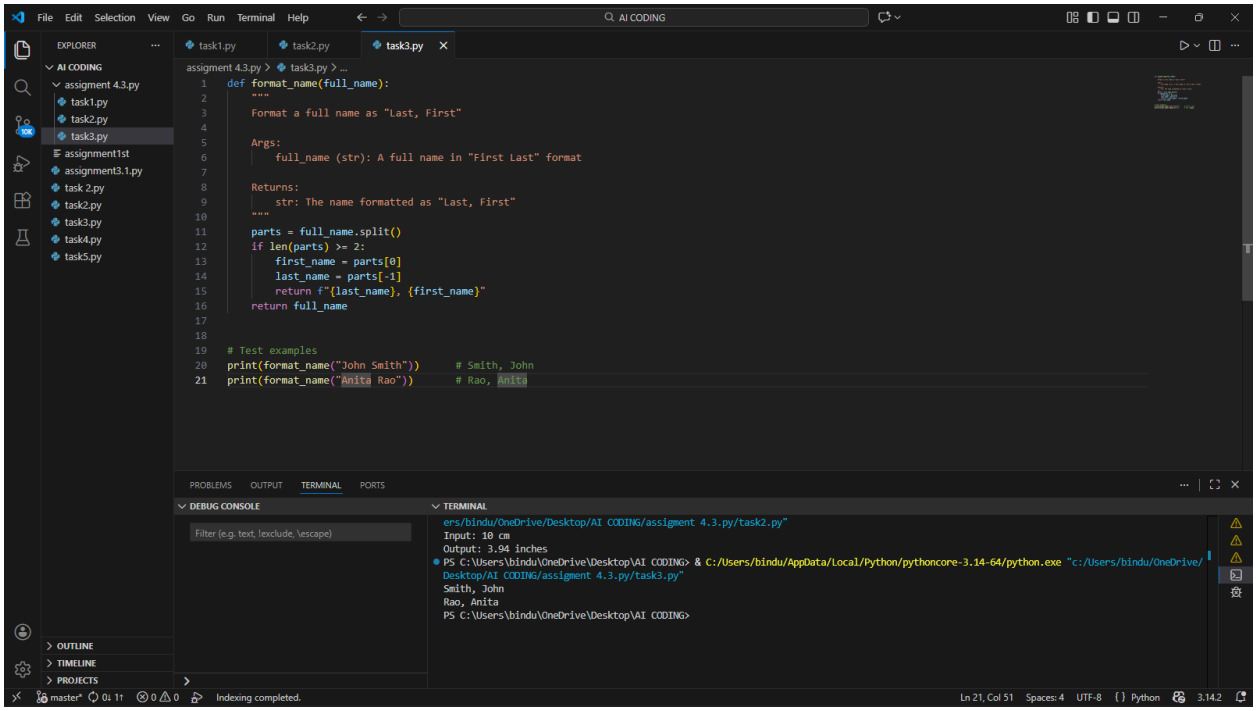
```
"""


defformat_name(full_name):
    """
    Format a full name as "Last, First"

    Args:
        full_name (str): A full name in "First Last" format

    Returns:
        str: The name formatted as "Last, First"
    """
    parts=full_name.split()
    iflen(parts) >=2:
        first_name=parts[0]
        last_name=parts[-1]
        returnf"{last_name}, {first_name}"
    returnfull_name


# Test examples
print(format_name("John Smith"))      # Smith, John
print(format_name("Anita Rao"))       # Rao, Anita
```

- **SampleInput:**

Enter full name: John Smith

- **SampleOutput:Smith, John**

ShortExplanationofLogic: The function splits the full name into first and last names and rearranges them in the required format it providing multiple examples improves few short



Task4:Comparative Analysis – Zero-Shot vs Few-Shot

- **Scenario:**
  Different prompt strategies may produce different code quality.
- **Prompt1:Zero-Shot Prompting**

Write a Python function that counts the number of vowels in a given string.
The function should return the total count.
Do not provide any examples.

Generatedcode:

```
def count_vowels(string):
    vowels = "aeiouAEIOU"
    count = 0
    for char in string:
        if char in vowels:
```

- **SampleInput:**
  Enter string: Hello World
- **SampleOutput:**

Number of vowels: 3

- **Prompt2:Few-Shot Prompting**

  - Write a Python function to count vowels in a string.
  - Examples:
  - Input: "hello" → Output: 2
  - Input: "AI Assisted Coding" → Output: 7

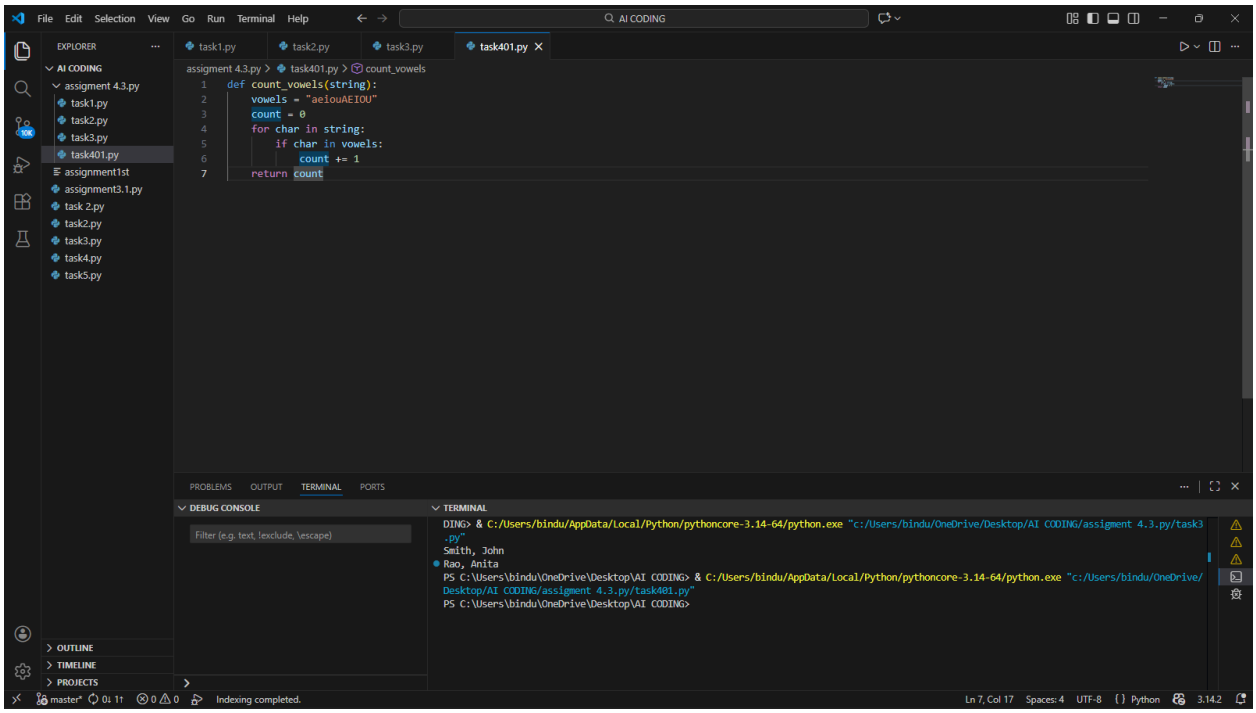
❖ **Generatedcode:**

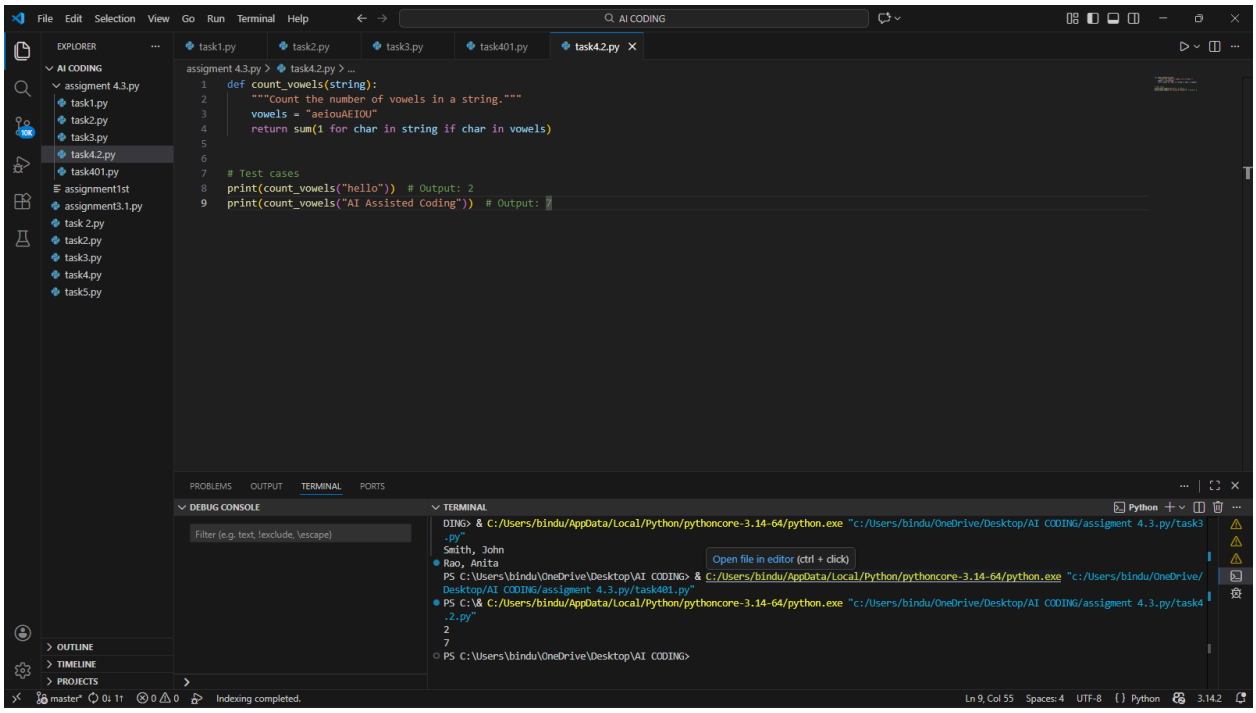
```
❖ efcount_vowels(string):
❖     """Count the number of vowels in a string."""
❖     vowels="aeiouAEIOU"
❖     returnsum(1forcharinstringifcharinvowels)
❖

❖ # Test cases
❖ print(count_vowels("hello"))  # Output: 2
❖ print(count_vowels("AI Assisted Coding"))  # Output: 7
```

  - **SampleInput:**
    Enter string: Hello World
  - **SampleOutput:3**

Explanation (Few-Shot);

The function uses a predefined vowel set and python's sum with the generator expression to count vowels efficiently the logic is compact and easier to understand due to example .





Comparison Table

Criteria	**Zero-Shot**	**Few-Shot**
Accuracy	Correct	Correct
Readability	Medium	High
Logical Clarity	Simple	Very clear
Code Length	Longer	Shorter

Task5:Few-Shot Prompting – File Handling (Line Count)

Scenario:You are building a text file analyzer that counts the number of lines in a file. Clear examples help ensure correct file handling.

Promptused:

 # Write a Python function to read a text file

 # The function should take the file name as input

 # Open the file in read mode# Count the total number of lines in the file

#Return the line count as an integer

 # Example:

 # If "file.txt" contains 3 lines, the output should be 3

Generatedcode:

```
defcount_lines(filename):
    """
    Read a text file and count the total number of lines.

    Args:
        filename (str): The name of the file to read

    Returns:
        int: The total number of lines in the file
    """
    try:
        withopen(filename, 'r') asfile:
            line_count=sum(1forlineinfile)
        returnline_count
    exceptFileNotFoundError:
        print(f"Error: File '{filename}' not found.")
        return0


# Example usage
if__name__=="__main__":
    result=count_lines("file.txt")
    print(f"Total lines: {result}")
```

- **SampleInput:**
- print(count_lines("file.txt"))
- **SampleOutput:**
  3

Short Explanation of Logic:
The program reads a text file in read mode and counts the number of lines present in it a counter variable is used to keep track of the number of lines. Each time a line is read from the filethe counter increases by one. After reading the entire file, the function returns the total number of lines. This method ensures accurate line counting and is easy to understand

File Edit Selection View Go Run Terminal Help

AI CODING

RUN AND DEBUG

RUN

Run and Debug

To customize Run and Debug create a launch.json file.

Debug using a terminal command or in an interactive chat.

Show automatic Python configurations

task1.py | task2.py | task3.py | task401.py | task4.2.py | task5.py

assigment 4.3.py > task5.py > ...

```python
def count_lines(filename):
    """
    Read a text file and count the total number of lines.

    Args:
        filename (str): The name of the file to read

    Returns:
        int: The total number of lines in the file
    """
    try:
        with open(filename, 'r') as file:
            line_count = sum(1 for line in file)
        return line_count
    except FileNotFoundError:
        print(f"Error: File '{filename}' not found.")
        return 0


# Example usage
if __name__ == "__main__":
    result = count_lines("file.txt")
    print(f"Total lines: {result}")
```

PROBLEMS OUTPUT TERMINAL PORTS

DEBUG CONSOLE

Filter (e.g. text, !exclude, \escape)

TERMINAL

```
PS C:\Users\bindu\OneDrive\Desktop\AI CODING> & C:/Users/bindu/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/bindu/OneDrive/Desktop/AI CODING/assigment 4.3.py/task5.py"
Error: File 'file.txt' not found.
Total lines: 0
PS C:\Users\bindu\OneDrive\Desktop\AI CODING>
```

BREAKPOINTS

- Raised Excepti...
- Uncaught Exce...
- User Uncaught...

master* 0↓ 1↑ ⊗ 0 ⚠ 0 Indexing completed. Ln 23, Col 36 Spaces: 4 UTF-8 { } Python 3.14.2